

matcami POS Sales

Business brief · what the till will do, how a sale flows, and what we need the business team to confirm · Plan approved for build on 28 September 2026 · Prepared from the technical plan tasks/pos-sales/

The next release lets the registered till take orders and payments. A cashier signs in with a PIN, opens the shift with a cash float, builds an order on a split screen (menu on the left, the customer's bill on the right), chooses how the customer is being served, takes cash or a card/mobile payment, and the sale is recorded, numbered, and posted to the books automatically. It keeps working when the internet drops. The owner sees the day's sales on the dashboard, by item and overall.

1. What the cashier will see

Step 1 · Sign in

Pick your name, type your PIN. Five wrong PINs lock you out for five minutes. The till returns to the sign-in screen after the idle time the owner set.

Step 2 · Open the shift

Enter the cash float in the drawer. The till shows the business date it will use for the shift. One shift is open at a time on a till.

Step 3 · Build the order

Menu categories are tabs; tap items to add them; choose extras (modifiers). The bill on the right never disappears. Pick **Sit now** (optionally a table number), **Waiting for a table**, or **Takeaway**.

Step 4 · Take payment

Choose Cash, Card or Mobile money. For cash, tap a quick amount or type what the customer handed over and the change is shown. Tap Pay. The screen shows the invoice number and the change.

• Online · 0 unsynced · Sam (Cashier) · • Session · business date 2026-09-28

Order type

Sit nowWaiting for a tableTakeaway

Table

4

Menu

BurgersDrinksSidesDesserts

Classic burger
\$8.00

Cheese burger
\$9.00

Veggie burger
\$8.50

Special (unavailable)

Extras panel opens inline: Extra cheese +\$0.50 · No tomato – \$0.00

Guest check

◇ 1 × Classic burger 8.00 · 10%8.00

+ Extra cheese0.50

◇ 2 × Cola 2.00 · 10%4.00

Subtotal12.50

Tax1.25

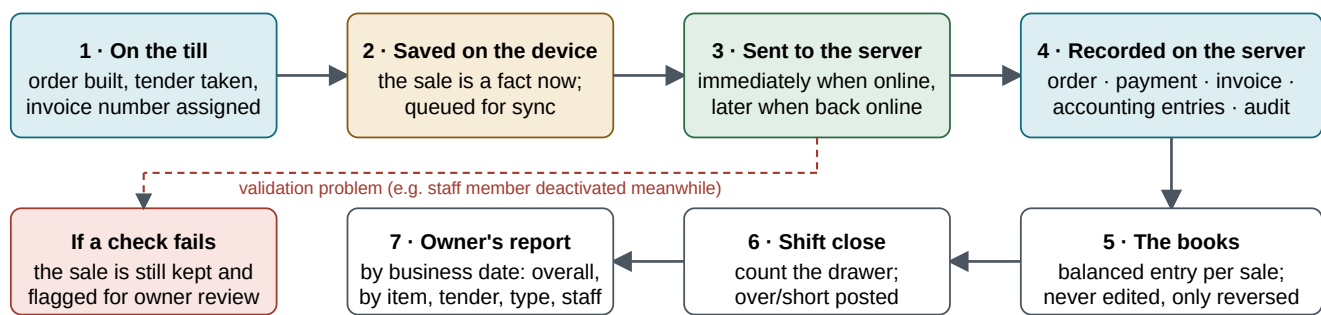
Total13.75

Pay

The order screen. The status bar is permanent: connection, unsynced count, who is signed in, and the shift's business date. Every status is a symbol as well as a colour so it reads under glare and for colour-blind staff.

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2. How a sale flows, end to end



Steps 1–2 work with no internet for cash sales. Card and mobile payments need the server's confirmation before they complete.

The workflow of one sale. The device is the first place a sale exists; the server is the source of truth for the books and reports.

The rules that make the numbers trustworthy

- **Prices and tax are frozen at the moment of sale.** Every line on a bill stores the price and the tax rate used. Changing a menu price or the tax setting tomorrow never changes yesterday's sales or reports.
- **One rounding rule.** Tax is worked out at full precision per line and rounded once on the bill total; the same function is used by the till, the server and the reports, so a receipt can never disagree with the books.
- **Every sale posts a balanced accounting entry** generated from a fixed rule table. Cash sales debit 1000 Cash on Hand; card and mobile sales debit the clearing accounts 1020 / 1030; revenue goes to 4000 Sales Revenue and tax to 2100 Tax Payable. Nobody types a debit or credit.
- **Posted records are permanent.** A paid order, an invoice, a payment or an accounting entry is never edited or deleted. A mistake is corrected by a reversing entry plus a new correct one, so the audit trail always shows what happened.
- **Money is whole cents.** No decimals, no floating point anywhere in money code.

3. The shift (POS session) and the business date

A shift starts with the opening float and ends with a count. The system computes the expected cash and posts any difference:

Item	Example	Where it comes from
Opening float	\$500.00	Typed at shift open
+ Cash sales	\$1,500.00	Recorded cash payments of this shift
= Expected cash	\$2,000.00	Computed by the server at close
Counted cash	\$1,990.00	Typed at shift close
Difference	-\$10.00	Posted to 6800 Cash Over/Short; zero posts nothing

Business date, not calendar date. A sale at 01:30 belongs to the shift that opened the evening before. Reports, the end of day and the drawer reconciliation all group by the shift's business date. The report page opens on the last shift's date by default so a visit at 00:40 is not empty.

Closing needs a connection and an empty sync queue, because the reconciliation runs on the server. The Close button is never dead: it says why it is blocked (offline, N operations still syncing). Refunds, pay-ins and pay-outs are not in this release, so they are zero in the formula.

4. When the internet drops

- **Cash sales keep working.** The sale is completed and numbered on the till and queued. The status bar always shows **Offline · N unsynced**. Staff can keep signing in with cached PINs.
- **Card and mobile are disabled while offline**, with the reason on the key, so nothing fails after the tap. (This is one of the decisions to confirm in section 7: the alternative is to treat a standalone terminal's approval as a fact and record it offline.)
- **Invoice numbers come from the till:** POS1-000001, POS1-000002... gap-free per device, online or offline. The server never rennumbers. A second terminal later gets its own prefix.
- **Sending twice is harmless.** Every operation carries a unique key; a retry after a lost response records nothing twice.
- **A sale is never rejected after the fact.** If a synced sale fails a check (for example the staff member was deactivated while the till was offline, or the price on the till was older than the current menu), it is still recorded and **flagged** for the owner. Only a tampered or unreadable operation is held back unrecorded; the shift cannot close until the owner retries or dismisses it, so cash in the drawer is never booked as an unexplained overage.
- **Sign-out and shift close are blocked while anything is unsynced**, and the till asks the browser to protect its storage, so unsynced sales survive a reload or a restart.

5. What the owner sees

The dashboard gains a **Reports** page for any business date: totals (gross sales, discounts, net sales, tax, takings, order count), sales by tender, by order type (dine-in / takeaway), by staff member, by item and by category, the day's shifts with opening, expected, counted and difference, and an alert when sales await review. A **Flagged sales** page lists each flagged operation with its details and two actions: **Retry** (record it now) or **Dismiss** with a reason. Both are audited. The dashboard home shows a notice when the count is above zero.

Report figures follow the accounting definitions: Gross sales – Discounts = Net sales; Net sales + Tax = Takings. Per-item rows may differ from the total by cents because of rounding, and the page says so.

6. What is not in this release, and why

Item	Status	Why
Receipt printing, kitchen tickets, cash drawer kick	next plan	Printing is a local print agent's job; it is a separate piece of work and never part of the payment step. Until it lands, the screen summary is the receipt.

Discounts, refunds, voids of sent items, comps, re-opening a paid order, pay-outs, opening the drawer without a sale	next plan	Each needs the owner's PIN approval and a reason code. The books and the rules for them are already prepared; the screens and approvals come next.
Table map, transferring or merging tables, splitting a bill, splitting a payment across methods	later	Kept simple: one bill, one tender, an optional table number. The data model already allows several payments per order.
Stock deduction and cost of goods sold	inventory plan	No recipes exist yet. The payment step already has the place where stock will be deducted; until then gross profit in the books equals revenue.
Home delivery	not built	Out of the agreed MVP scope. "Delivery" in the original request is treated as Takeaway; adding delivery later is one small change.
Kitchen "send" button, waiter handhelds, second terminal	later	The design leaves room for them; nothing has to be rebuilt.

7. Decisions we assumed — please confirm or change

The plan was written with the defaults below. Any change is fine now; some become costly once sales exist.

#	Question	Default in the plan
1	"Delivery": did you mean takeaway (customer collects) or home delivery (addresses, drivers)?	Takeaway. Home delivery is a later plan.
2	Finish the employee-roles work (PR #11) before starting this build?	Yes. This plan is written on top of it.
3	Which tenders at launch?	Cash, Card and Mobile money are all built as "recorded from a standalone terminal" tenders; the owner switches Card and Mobile on or off in Settings. Cash is always on.
4	Card and mobile when the internet is down?	Disabled, with the reason shown. Alternative: treat the terminal's approval as a fact and record it offline.
5	Create all 23 accounts of the chart for every restaurant, or only the clearing accounts for accepted tenders?	All 23, so switching a tender on later needs no accounting change.
6	Who may open and close a shift?	Anyone with the "take payment" permission (cashiers and the owner). No new permission is introduced.
7	Rounding detail	Total, tax and discount are each rounded once; net and subtotal are derived from them, so every entry balances to the cent.
8	Report scope for this release	Sales, shifts and flagged sales. Journal and trial balance views come with a later reporting plan.
9	More than one open order at a time on the till?	No: one order at a time, paid at the counter.
10	Return to the sign-in screen after every payment?	No; only after the idle time.
11	A "Send to kitchen" button before printing exists?	No; the only closer on the order screen is Pay.
12	Should removing an item from an unpaid bill be logged?	No. Removing a not-yet-sent item needs no reason; the drawer count is the control.
13	How long does the till keep synced sales locally?	30 days, then pruned; the server keeps everything.

8. Risks the review found, and how the plan handles them

The plan went through a five-lens review (accounting, data model, offline, security, operations) with an adversarial pass. Of 49 findings, 24 were refuted; the four most serious survivors, in plain terms:

- **A flagged sale must never vanish into a "to review" pile that the books ignore.** Otherwise its cash would show up as an unexplained overage at shift close. The plan records such sales in full and flags them, and blocks the close while anything is genuinely unrecorded.
- **Tax-inclusive pricing can produce an entry that is one cent out of balance** if net and tax are rounded separately (for example 20% on a \$9.99 item). The plan derives the figures so the entry balances by construction and tests every rate in both tax modes.

- **The sync endpoint must not be a back door for unapproved discounts or refunds.** The server accepts only what this release's screens can produce: no discounts, positive quantities, prices that match the menu the till had.
- **Replacing a till (revoke and re-register) must not strand or misfile the sales still on it.** Every queued sale carries the device it was made on; the server files it under that device, and any live till of the restaurant may close the old till's shift.

9. Build plan at a glance

The technical plan is a directory of eleven files: an overview and one file per feature area, 41 tasks in seven phases, each task with its own tests. Money, accounting and offline behaviour carry mandatory automated tests before any screen is built.

Phase	Feature file	What it delivers
0	Decisions	Verify the base, record the decisions above, define the sale message format shared by till and server
1	Schema	Orders, payments, invoices, shifts, sync log, chart of accounts, journal; the database-enforced balance rule
2	Money · Accounting · Orders	Order totals in both tax modes; the posting rules; the single payment transaction; shift open/close; the sync handler
3	Offline queue	The till's local order store, invoice numbering, queue and retry rules, signed-in state and idle lock
4	API	The sync endpoint with permission checks; tender settings
5	Till screens · Dashboard	Shift, order and payment screens; the Reports and Flagged sales pages
6	Verification	End-to-end journeys (a full shift online and offline), documentation, the final gate

What we need from the business team: confirm or change the thirteen defaults in section 7, and tell us whether your card terminals are standalone devices (this decides question 4). Everything else in this brief is already specified in the plan.